time machine. Instead, we dealt *only* with simple, inanimate time-traveling things, such as electromagnetic waves.

Before publishing, we thought a lot about waves that travel backward in time through a wormhole; we searched hard for unresolvable paradoxes in the waves' evolution. Ultimately (and with crucial proddings from John Friedman), we convinced ourselves that there probably will be *no unresolvable paradoxes*, and we conjectured so in our paper.[12] We even broadened our conjecture to suggest that there would never be unresolvable paradoxes for *any* inanimate object that passes through the wormhole. It was this conjecture that created the most controversy.

Of the letters we received, the most interesting was from Joe Polchinski, a professor of physics at the University of Texas in Austin. Polchinski wrote, "Dear Kip, . . . If I understand correctly, you are conjecturing that in your [wormhole-based time machine there will be no unresolvable paradoxes]. It seems to me that . . . this is not the case." He then posed an elegant and simple variant of the matricide paradox—a variant that is *not* entangled with free will and that we therefore felt competent to analyze:

Take a wormhole that has been made into a time machine, and place its two mouths at rest near each other, out in interplanetary space (Figure 14.9). Then, if a billiard ball is launched toward the right mouth from an appropriate initial location and with an appropriate initial velocity, the ball will enter the right mouth, travel backward in time, and fly out of the left mouth before it entered the right (as seen by you and me outside the wormhole), and it will then hit its younger self, thereby preventing itself from ever entering the right mouth and hitting itself.

This situation, like the matricide paradox, entails going back in time and changing history. In the matricide paradox, I go back in time and, by killing my mother, prevent myself from being born. In Polchinski's paradox, the billiard ball goes back in time and, by hitting itself, prevents itself from ever going back in time.

Both situations are nonsensical. Just as the laws of physics must be logically consistent with each other, so also the evolution of the Universe, as governed by the laws of physics, must be fully consistent with itself—or at least it must be so when the Universe is behaving classically (non–quantum mechanically); the quantum mechanical realm is a little more subtle. Since both I and a billiard ball are highly classical objects (that is, we can exhibit quantum mechanical behavior only when one makes